

COTONOU, Benin

WMO: 653440

Lat: 6.357N

Lon: 2.384E

Elev: 19

StdP: 14.69

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	71.7	73.1	55.0	64.3	83.5	62.3	84.1	80.0	20.4	80.5	18.9	80.5	5.7	20	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.5	91.4	80.4	90.3	80.7	89.8	80.7	84.1	88.8	83.2	87.8	82.6	87.2	11.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	170.2	87.6	82.1	167.4	87.2	81.0	160.9	86.0	48.1	89.6	47.3	88.6	46.6	88.1	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.0	16.5	15.6	DB	68.3	94.2	3.1	2.0	66.1	95.6	64.3	96.8	62.6	98.0	60.4	99.4
			WB	60.7	84.9	4.2	2.0	57.7	86.4	55.2	87.6	52.8	88.7	49.7	90.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	82.5	84.5	85.0	84.2	83.0	80.6	79.3	78.7	79.8	81.0	83.1	83.2
	DBStd	2.75	2.02	1.88	2.14	2.42	2.36	2.02	1.52	1.29	1.45	1.78	1.52	1.62
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11700	1008	965	1084	1026	1022	918	910	891	893	961	994	1028
	CDD65	6225	543	545	619	576	557	468	445	426	443	496	544	563
	CDH74	67800	6044	6786	7969	7261	6683	4675	3619	2972	3767	5008	6432	6584
	CDH80	23331	2184	2927	3666	3212	2640	1144	439	286	575	1316	2410	2532
Wind	WSAvg	8.9	7.3	9.4	10.2	9.3	7.8	8.4	11.0	11.3	10.1	7.8	7.4	7.1
Precipitation	PrecAvg	37.6	0.5	1.0	1.9	4.3	6.7	9.6	3.7	0.9	4.1	4.3	0.9	0.3
	PrecMax	57.4	2.6	4.7	4.6	12.7	15.9	17.5	17.7	4.3	15.1	10.6	3.0	1.7
	PrecMin	21.4	0.0	0.0	0.6	0.3	2.9	5.2	0.8	0.1	0.4	0.2	0.0	0.0
	PrecStd	10.2	0.7	1.3	0.9	3.0	3.2	4.0	3.7	1.1	3.3	2.8	0.8	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.4	91.8	91.8	91.7	91.4	88.3	85.7	84.5	86.2	88.2	90.5	91.6
		MCWB	72.0	79.3	81.5	82.3	81.8	79.9	78.7	77.3	78.7	80.1	79.5	75.9
	2%	DB	89.7	90.4	91.2	91.0	89.9	87.5	84.3	83.5	84.8	87.7	89.7	89.9
		MCWB	77.9	80.7	82.1	82.2	81.2	79.8	77.8	76.9	78.1	80.1	80.4	78.8
	5%	DB	88.2	89.7	89.9	89.8	89.3	86.1	83.4	82.6	84.2	86.4	89.1	89.4
		MCWB	78.9	81.0	81.5	81.6	81.0	79.3	77.4	76.5	77.8	79.4	80.5	79.4
	10%	DB	87.6	89.1	89.5	89.3	88.0	84.7	82.5	82.0	83.1	85.8	88.0	88.1
		MCWB	79.0	80.9	81.4	81.4	80.6	78.7	77.0	76.3	77.4	79.2	80.4	79.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.6	84.6	84.6	84.6	83.7	81.8	79.9	79.1	80.0	81.9	83.1	83.6
		MCDB	87.4	88.7	89.8	89.4	88.4	85.4	83.1	82.1	83.4	85.9	87.8	87.8
	2%	WB	82.4	83.7	83.8	83.6	82.5	80.7	79.0	78.2	79.2	80.8	82.4	82.5
		MCDB	86.3	87.9	89.1	88.6	87.7	85.0	82.4	81.4	83.0	85.6	87.3	87.0
	5%	WB	81.5	82.6	82.7	82.7	81.7	79.9	78.4	77.4	78.6	80.1	81.7	81.6
		MCDB	85.5	86.9	87.6	87.5	87.0	84.2	81.7	80.8	82.6	84.8	86.7	86.2
	10%	WB	80.8	81.9	82.0	81.9	81.0	79.2	77.8	76.9	78.1	79.4	80.8	80.8
		MCDB	84.8	86.3	86.8	86.8	86.3	83.4	81.1	80.4	81.8	83.9	85.9	85.5
Mean Daily Temperature Range	5% DB	MDBR	11.7	10.0	9.5	9.8	10.1	8.5	6.8	6.9	7.7	9.4	10.7	11.8
		MCDBR	12.4	10.3	9.6	9.8	10.3	9.8	7.5	7.6	8.6	10.3	11.3	12.7
		MCWBR	6.1	5.0	5.0	5.0	5.1	4.9	3.8	3.7	4.1	5.0	5.1	5.5
	5% WB	MCDBR	10.1	9.3	9.4	9.5	10.0	9.1	7.1	7.1	8.2	9.9	10.6	10.7
MCWBR		5.2	5.1	5.5	5.7	5.7	5.3	4.2	3.9	4.5	5.3	5.7	5.4	
Clear-Sky Solar Irradiance	taub	0.834	0.840	0.827	0.662	0.555	0.524	0.529	0.532	0.489	0.509	0.603	0.708	
	taud	1.499	1.500	1.515	1.808	2.042	2.146	2.128	2.110	2.262	2.196	1.899	1.694	
	Ebn at Noon	181	186	190	221	240	245	245	248	264	259	231	204	
	Edh at Noon	95	97	96	70	54	48	49	51	45	48	63	77	
All-Sky Solar Radiation	RadAvg	1576	1573	1646	1661	1508	1253	1323	1361	1415	1535	1619	1574	
	RadStd	55	83	84	69	76	85	106	89	71	49	46	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.50	+0.56	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.50	+0.56	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page